Haomin (Harmin) Qi

harminchee.github.io

Interests

Open to roles in **Applied/Research Scientist**, **Machine Learning Engineer**, **AI/Agent Developer**, **Software Engineer**, and broader **IT roles** (technical support, platform, data). Technical interests: coding agents and multi-agent planning, LLM inference and training, multimodal, LLM security and safety-aware code generation.

Education

University of California San Diego Sep. 2025 – Nov. 2026
M.S. in Electrical and Computer Engineering La Jolla, CA
GPA: 3.8 /4.0 | Track in Machine Learning and Data Science

The Chinese University of Hong Kong Sep. 2021 – Jul. 2025
B.S. in Mathematics and Information Engineering Hong Kong SAR
Double Major Graduation | Elite Stream Class

University of Leeds Jan. 2024 – Jul. 2024
Abroad Exchange in Computer Science Leeds, UK
GPA: 4.00 /4.00 | First Class Honour

Experience

Shang Data Lab, UC San Diego Sep. 2025 – Present
Research Assistant | Supervisor: Jingbo Shang La Jolla, CA

- Developed **PoCGYM**, a benchmark construction framework for realistic bug injection into production Java repositories. Grounded injection in real execution traces and bug-pattern specifications, with optional semantic transform stages (call-stack deepening, aliasing, predicate reframing) to control artifact difficulty and realism
- Built an end-to-end pipeline spanning injection, Maven/Surefire validation, and archival; defined two agent tasks—**Trigger** (write an exposing test) and **Detect** (localize bug from a pattern description)—on real-code bugs

Advanced Wireless Systems Group, CUHK Apr. 2024 – Sep. 2025
Research Assistant | Supervisor: Soung Chang Liew Hong Kong SAR

- Led **VeriRAG**, the first RAG framework for RTL DFT compliance repair; trained a contrastive autoencoder for structure-aware Verilog retrieval with an iterative Xcelium+LEC revision loop, achieving **7.72**× repair rate over zero-shot GPT-o1
- Designed **TopoEdge**, an edge agentic framework for SDN configuration generation/repair; trained a contrastive GCN over 1,879 FRRouting topology graphs, grounding a three-agent generate–verify–repair loop on a Pi cluster at **Pass@20=0.890**

Deloitte Sep. 2024 – Dec. 2024
Machine Learning Developer Intern Hong Kong SAR

- Built **ABA-RAG**, fusing multimodal IoT signals (BVP, GSR, temperature, acceleration) via a Transformer classifier (**73% acc., 0.90 recall**) with a three-layer RAG pipeline using LoRA-GA fine-tuned Llama-3.1-70B, scoring **9.3/10** among four frontier LLMs
- Deployed as a web platform with task generation, therapy management, and IoT dashboards; pilot across N=10 learners yielded expert ratings of **9.63/10** (ICC=0.76), on par with practitioner-led interventions

Wireless Ad-Hoc & Sensor Networks Lab, NCU Jun. 2024 – Sep. 2024
Research Intern | Supervisor: Min-Te Sun Taoyuan, TW

- Proposed **Hybrid PEFT**, fusing LoRA-GA and BOFT per layer via gradient-norm-based adaptive mixing; reduced training time by **2.1**× and peak GPU memory by **50%** across 7B–405B models, outperforming individual baselines on GLUE, GSM8K, MT-Bench, and HumanEval
- Pioneered **uRNN**-adapted transformer fine-tuning by embedding structured unitary matrices into attention and FFN layers with manifold-preserving gradient updates; achieved consistent gains over LoRA and BOFT on XNLI and FLORES

Intell-Pro Global | R-Guardian May. 2023 – Present
Co-Founder & AI Lead Hong Kong SAR

- Architected a trademark similarity engine combining CNN-based image feature extraction, template matching, and reverse image search over scalable cloud infrastructure for real-time retrieval across multiple national IP databases
- Co-founded **Intell-Pro Global Limited**, serving law firms and IP agencies across the US, Europe, and Asia; secured **HK$675,000** in TSSSU funding and the **HK Tech300 Entrepreneurship Award**

Artificial Intelligence & Computer Vision Lab, NYCU Jun. 2023 – Aug. 2023
Research Intern | Supervisor: Jun-Wei Hsieh Taipei, TW

- Contributed to **DeepMAD**, a training-free NAS framework formulating CNN design as entropy maximization subject to effectiveness constraints; validated on ImageNet, achieving **82.8% top-1 accuracy** at **50% fewer FLOPs** vs. conventional baselines

PUBLICATIONS

PoCGYM: Evaluating AI Coding Agents for PoC Generation via Verifiable Bug Injection
Haomin Qi, Xiangzhe Xu, Yiming Huang, Jingbo Shang, Chengpeng Wang
Under review at Neural Information Processing Systems ***(NeurIPS 26')***

Topology-Grounded Agentic Framework for Edge Networking Code Generation and Repair
Haomin Qi, Bohan Liu, Zihan Dai, Yunkai Gao, Soung Chang Liew
IEEE Wireless Communications and Networking Conference ***(IEEE WCNC 26')***

VeriRAG: A Retrieval-Augmented Framework for Automated RTL Testability Repair
Haomin Qi*, Yuyang Du*, Lihao Zhang, Soung Chang Liew, Kexin Chen, Yining Du
International Symposium on Quality Electronic Design ***(ISQED 26')***

GraphCue for SDN Configuration Code Synthesis
Haomin Qi, Fengfei Yu, Chengbo Huang
IEEE Consumer Communications & Networking Conference 2026 ***(IEEE CCNC 26')***

Governance-Aware Hybrid Fine-Tuning for Multilingual Large Language Models
Haomin Qi, Chengbo Huang, Zihan Dai, Yunkai Gao, Min-Te Sun
IEEE International Conference on Big Data 2025 Workshop LLM4All ***(IEEE BigData 25'LLM4All)***

Hybrid and Unitary PEFT for Resource-Efficient Large Language Models
Haomin Qi, Zihan Dai, Chengbo Huang
American Journal of Computer Science and Technology ***(AJCST)***

Transforming ABA Therapy: An IoT-Guided, Retrieval-Augmented LLM Framework
Haomin Qi, Chung-Ho Sin, Rosanna Yuen-Yan Chan, Victor Chun-Man Wong
IEEE Access

PROJECTS

DAP: Panoramic Metric Depth Foundation Model

- Built a panoramic metric depth foundation model on a **2M-sample data engine** (UE5 AirSim360, Structured3D, 1.7M real panoramas) via a three-stage semi-supervised pipeline with Scene-Invariant and Realism-Invariant labelers bridging synthetic/real domain gaps
- Proposed a DINOv3-Large backbone with plug-and-play range mask heads and distortion-aware multi-loss optimization; achieved SOTA zero-shot metric depth on Stanford2D3D, Matterport3D, and Deep360

MWRCNN: Multi-Stage Wavelet Dynamic CNN for Image Restoration

- Extended MWDCNN to a unified restoration framework (deblurring, JPEG artifact removal, super-resolution, low-light, mixed degradation) by redesigning the Dynamic Convolutional Block's Weight Generator to preserve spatial context beyond global average pooling
- Outperformed TV and MWDCNN on BSD68: >**3.5 dB PSNR** at σ=50, >**0.5 dB** on GoPro real deblurring, and **50% LPIPS reduction** under mixed degradation

Adaptive Test-Time Scaling for LLM Safety Guards

- Studied parallel, sequential, and latent test-time scaling for GuardReasoner on WildJailbreak; led parallel scaling via DeepConf confidence-weighted aggregation, raising F1 to **0.759** at k=8, outperforming majority voting at all sample budgets
- Designed confidence-triggered adaptive policies; a two-level policy at τ=0.8 achieved **92% accuracy at 63% of peak compute** vs. 91% static full-budget baseline; diffusion latent scaling reached **91% accuracy** at 50 inference steps

SKILLS

Languages: Python, Java, C, SQL, Bash, HTML, JavaScript/TypeScript, Verilog, P4
LLM Training: PyTorch, Hugging Face Transformers, PEFT, LoRA//LoRA-GA/BOFT, Accelerate, DeepSpeed, vLLM
Agents: LangGraph, LangChain, Claude Code, Codex, OpenAI API, Anthropic API, AWS Bedrock, RAG
Tools & Infra: Maven/Surefire, JUnit, Pytest, Docker, Git, Linux, FastAPI, MLflow, Weights & Biases, CI/CD